Koneru Lakshmaiah Education Foundation
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Off-Campus: Bachupally-Gandimaisamma Road, Bowrampet, Hyderabad, Telangana - 500 043.
Phone No: 7815926816, www.klh.edu.in

MACHINE LEARING-(25SC2107E)

Y25 - 2026-2027 Trimester - 4

Project Abstract Submission Form

Sec No-	Team No-	**Project Title-** Chronic Kidney Disease Prediction Using XGBoost and LightGBM Gradient-Boosted Trees

1. Abstract

Chronic Kidney Disease (CKD) is a progressive medical condition that can lead to severe complications if it is not detected and managed at an early stage. Early prediction of CKD can support timely diagnosis and appropriate treatment. This project proposes a machine learning-based approach for predicting Chronic Kidney Disease using XGBoost and LightGBM, two efficient gradient-boosted tree algorithms. The proposed system uses clinical and laboratory features such as age, blood pressure, specific gravity, albumin, blood glucose, serum creatinine, hemoglobin, and other relevant health indicators to classify patients as having CKD or not. The dataset is preprocessed by handling missing values, encoding categorical attributes, and selecting relevant features before model training. Both XGBoost and LightGBM models are trained and evaluated using performance metrics such as accuracy, precision, recall, F1-score, and ROC-AUC. The performance of the two models is compared to identify the more effective approach for CKD prediction. The proposed system aims to provide an accurate and efficient predictive model that can assist healthcare professionals in early CKD risk identification and support data-driven clinical decision-making.

2. Team Details

Roll Number(s)	Name of the Student	Signature
2520030170	**Abhiram**	
2520030299	**Shaila**	

3. Frameworks / Platforms / Software Used

- **Python** – Primary programming language for data preprocessing, model development, and evaluation.
- **Pandas** – Used for data manipulation, cleaning, and preprocessing.
- **NumPy** – Used for numerical computations and handling datasets.
- **Scikit-learn** – Used for data preprocessing, train-test splitting, feature scaling, and model evaluation.
- **XGBoost** – Used to build the XGBoost gradient-boosted tree classification model.
- **LightGBM** – Used to develop the LightGBM gradient-boosted tree classification model.
- **Matplotlib / Seaborn** – Used for data visualization and graphical representation of model performance.
- **Jupyter Notebook / Google Colab** – Used as the development and experimentation environment.
- **VS Code** – Optional code editor for developing and managing the project.
- **CSV Dataset** – Used as the primary data source containing clinical attributes for CKD prediction.

Guide Name and Signature (remarks if any) _____